

Objective

To implement and demonstrate Multiple Inheritance using Interfaces in Java, where class C1 implements both I1 and I2 interfaces and accesses their variables and methods.

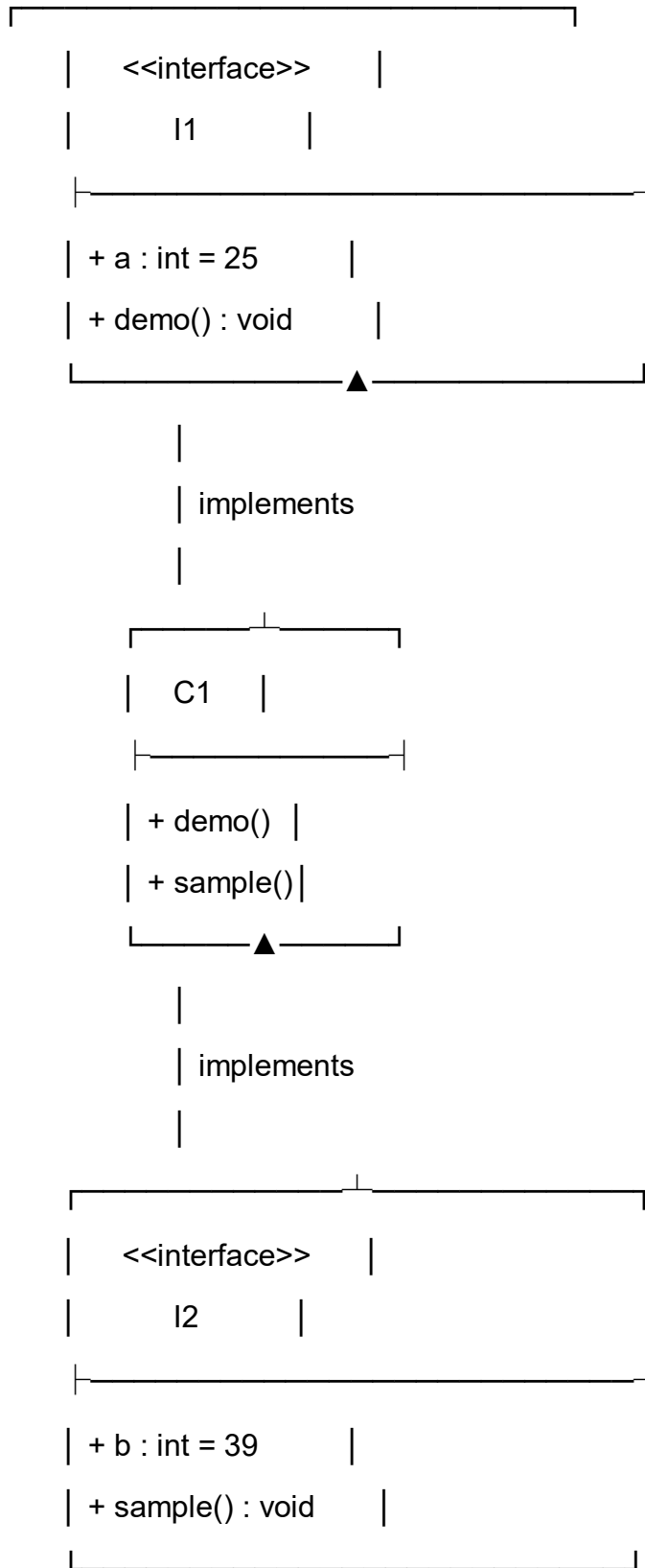
Steps for Execution

1. Create a Java project and package named javacore.
2. Create a file named C1.java and enter the given code.
3. Define interfaces I1 and I2 with variables and abstract methods.
4. Create class C1 implementing both interfaces.
5. Implement the demo() and sample() methods.
6. Create an object of C1 in the main() method.
7. Display the interface variables and call both methods.
8. Compile and run the program.
9. Observe the output.

Algorithm

1. Start the program.
2. Define interface I1 with variable a and method demo().
3. Define interface I2 with variable b and method sample().
4. Create class C1 implementing I1 and I2.
5. Implement demo() and sample().
6. Create a C1 object.
7. Display a and b.
8. Call demo() and sample().
9. Display the output.
10. Stop the program.

UML CLASS DIAGRAM:



Relationships

- C1 implements I1
- C1 implements I2
- C1 provides implementations for demo() and sample().
- Interface variable a has value 25.
- Interface variable b has value 39.

CODE:

```
package javacore;
```

```
interface I1{
```

```
    int a=25;
```

```
    void demo();
```

```
}
```

```
interface I2{
```

```
    int b=39;
```

```
    void sample();
```

```
}
```

```
public class C1 implements I1,I2{
```

```
    @Override
```

```
    public void demo() {
```

```
        System.out.println("this is the abstract method of interface1");
```

```
    }
```

```
    @Override
```

```
    public void sample() {
```

```
        System.out.println("this is the abstract method of interface 2");
```

```
    }
```

```
public static void main(String[] args){  
    C1 ac=new C1();  
    System.out.println(ac.a);  
    System.out.println(ac.b);  
    ac.demo();  
    ac.sample();  
  
    }  
}
```

OUTPUT:

25

39

this is the abstract method of interface1

this is the abstract method of interface 2

Result:

The program successfully demonstrates Multiple Inheritance using Interfaces in Java. The class C1 implements both I1 and I2, accesses their variables, and provides implementations for their abstract methods.